



H2 Biology

Paper 1 Multiple Choice

9744/01

20 September 2018

60 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use paper clips, glue or correction fluid.

Write your name, civics group and registration number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

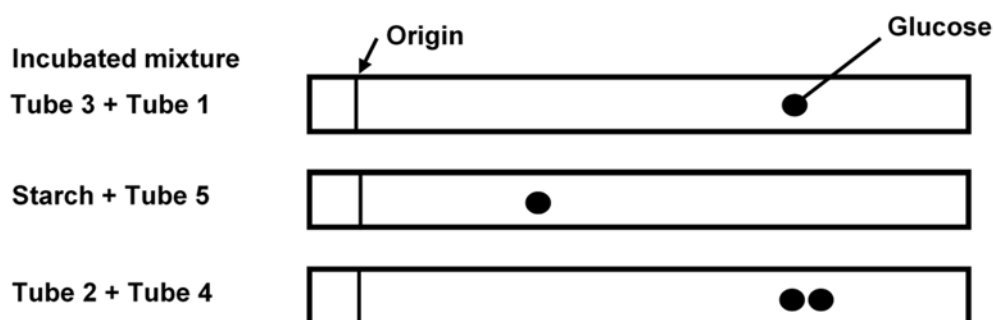
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **23** printed pages and **1** blank pages.

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- 1 A lab technician has 5 tubes containing unknown compounds. However, he knows that each tube contains only either a carbohydrate or enzyme. To determine the identities of the compounds in the 5 tubes, the lab technician conducted various tests and he made the following observations.

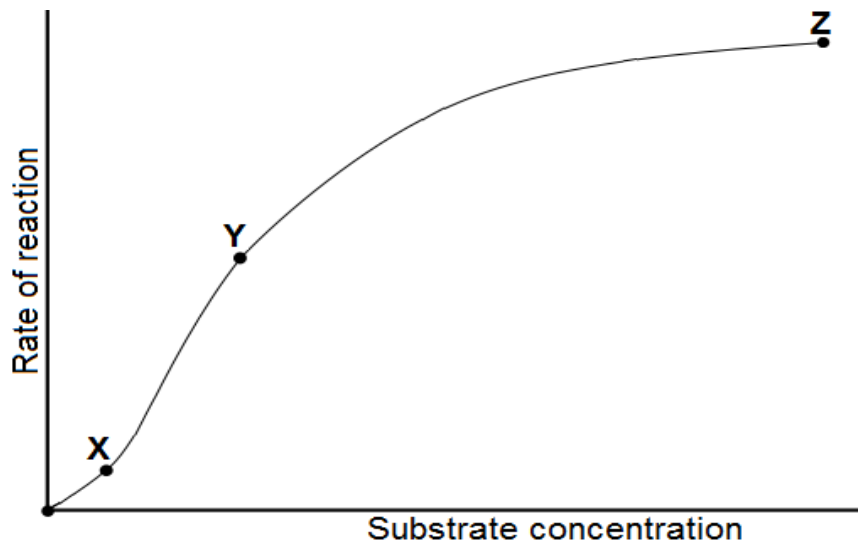
- I Sample in tube 2 is soluble in water. Sample in tube 3 was insoluble in water.
- II Tube 5 tested positive with Biuret's test.
- III Contents of all 5 samples tested negative with Benedict's test.
- IV However, after he mixed and incubated the contents of certain tubes together, a positive result was obtained for Benedict's test. The mixtures were also subjected to paper chromatography, and the results were shown below.



Which of the following correctly shows the contents of each tube?

	tube 1	tube 2	tube 3	tube 4	tube 5
A	amylase	cellulose	sucrose	cellulase	sucrose
B	cellulase	sucrose	cellulose	sucrase	amylase
C	cellulose	sucrose	cellulose	sucrase	amylase
D	sucrose	cellulose	sucrase	amylase	cellulase

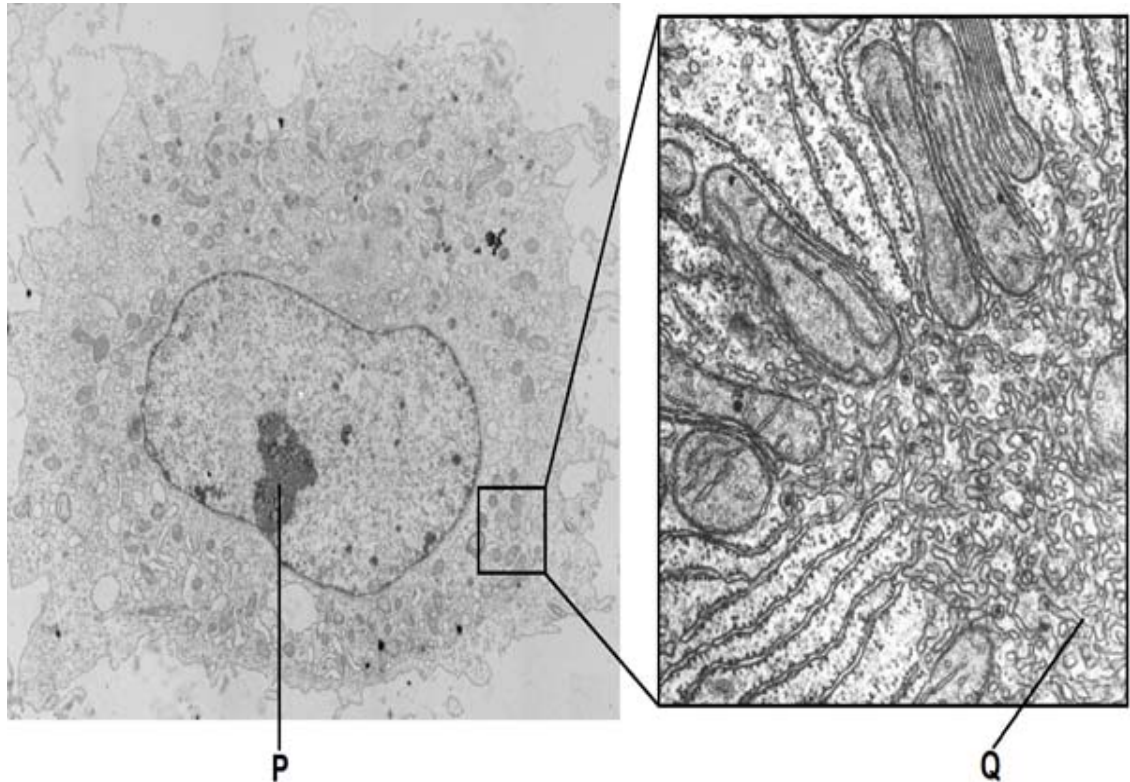
- 2 The graph below shows the effect of increasing substrate concentrations on the activity of an allosteric enzyme under optimum conditions.



Which of the following statements is correct?

- A There is low kinetic energy at **X** to overcome the activation energy, thus resulting in a low rate of reaction.
- B Rate of reaction increases at a faster rate at **Y** as the allosteric activator outcompetes the allosteric inhibitor to bind to the allosteric site.
- C At **Z**, enzyme molecules are in the active state and active sites are saturated.
- D Substrate concentration is the limiting factor at **X** and **Y** but temperature is the limiting factor at **Z**.

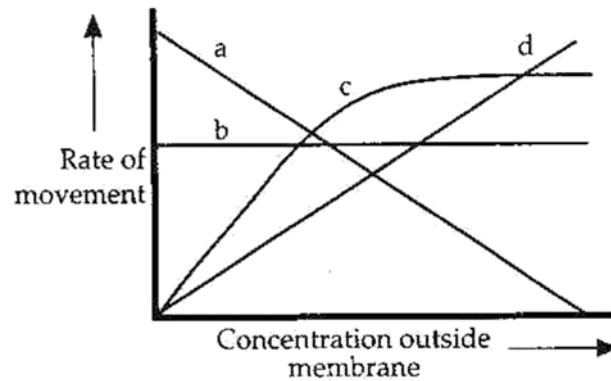
- 3 The diagram on the left below shows the electron-micrograph of a cell found in a healthy individual. A region of the cell, indicated by the box, is magnified and shown in the diagram on the right.



Which of the following shows the correct identities and functions of **P** and **Q**?

	Identity of P	Function of P	Identity of Q	Function of Q
A	Nucleosome	Transcription of rRNA	Smooth ER	Detoxification
B	Nucleosome	Assembly of ribosomal subunits	Secretory vesicles	Storage of Ca^{2+}
C	Nucleolus	Assembly of ribosomal subunits	Smooth ER	Synthesis of lipids
D	Nucleolus	Transcription of rRNA	Secretory vesicles	Modification of protein

- 4 Which of the following curves best represent the movement of substances by simple diffusion and facilitated diffusion?



	simple diffusion	facilitated diffusion
A	d	b
B	d	c
C	c	a
D	c	d

- 5 The figures below show the complete karyotypes of 2 rodents of the same species. In this species of rodents, males are heterogametic.



Rodent A (Male)

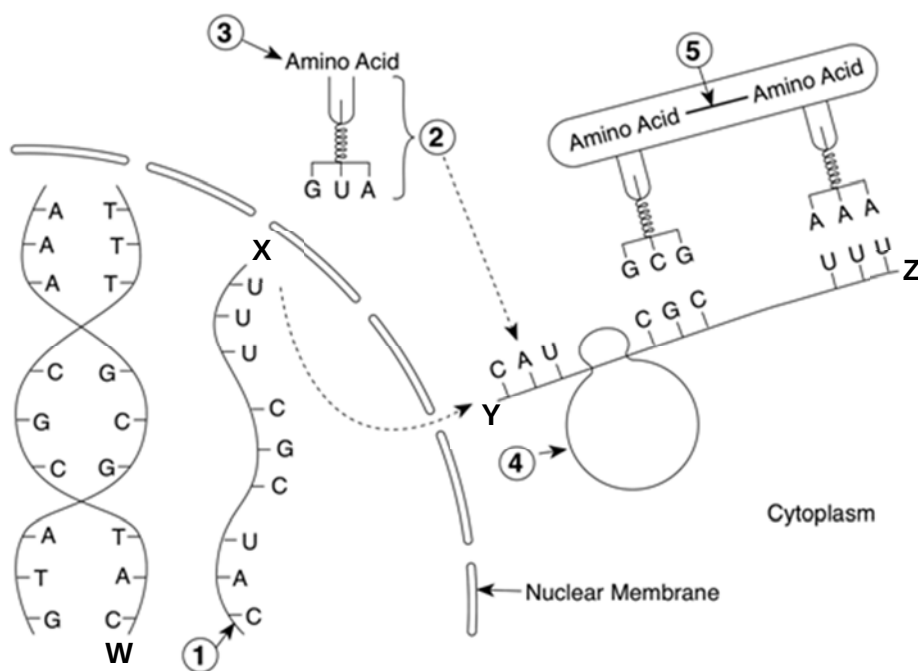


Rodent B (Female)

Which of the following observations is not true?

- A** A chromosomal aberration occurred in the ovary of the mother of rodent **B**.
- B** Rodent **A** is diploid and $2n = 16$.
- C** Rodent **B** has 1 missing X chromosome.
- D** Non-disjunction of autosomes occurred in rodent **B**.

- 6 The diagram below represents some biochemical reactions involved in protein synthesis.



Which is correct?

	Entire molecule coded directly from DNA is represented by	5' end of molecule	Enzyme involved in catalysing bond 5
A	1 and 2	Z	peptidyl transferase
B	1 and 2	Y	aminoacyl tRNA synthetase
C	1, 2 and 3	X	aminoacyl tRNA synthetase
D	1, 2 and 4	W	peptidyl transferase

- 7 A mutation of a gene coding for an ion pump in cell surface membranes results in the substitution of one amino acid, arginine, by another, histidine.

The DNA triplet codes for the two amino acids are shown.

Arginine GCA
GCG
GCT
GCC
TCT
TCC

Histidine GTA
GTG

Which mutation has occurred in the DNA?

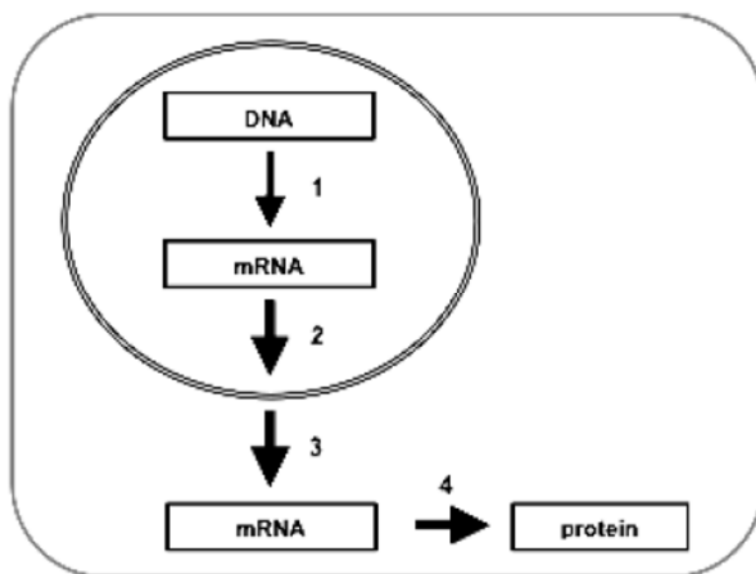
- A Addition of an extra nucleotide
- B Replacement of a nucleotide with a purine base by one with a different purine base
- C Replacement of a nucleotide with a purine base by one with a pyrimidine base
- D Replacement of a nucleotide with a pyrimidine base by one with a different pyrimidine base
- 8 The electron micrograph shows part of a bacterial cell. The DNA molecule has been identified. Identify structures **P** and **R** as well as the process(es) that is/are occurring below.



	Structure P	Structure R	Process(es)
A	DNA polymerase	Daughter DNA	DNA replication
B	String of nucleosomes	mRNA	Transcription
C	Polyribosomes	Polypeptides	Translation
D	Polyribosomes	mRNA	Transcription and translation

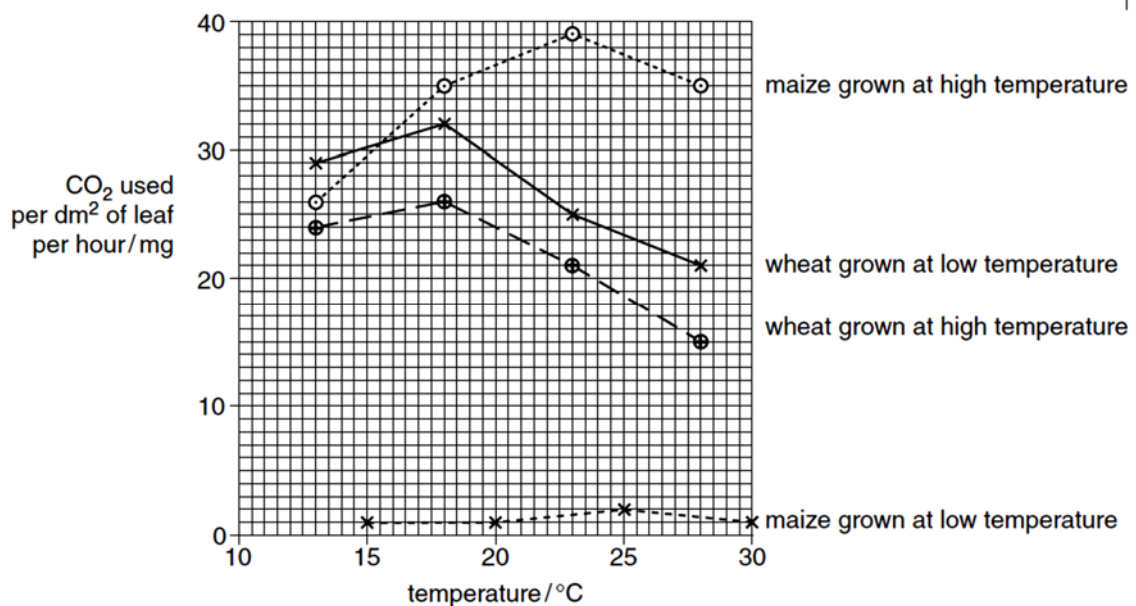
- 9 The following diagram shows the expression of a particular gene to its protein product in a eukaryotic cell.

Which combination correctly describes steps 1 – 4?



	1	2	3	4
A	DNA is demethylated	Alternative splicing occurs	Activators bind to enhancers	Ribosome binds to 5' UTR
B	DNA is demethylated	5' capping occurs	RNase does not degrade 5' capped mRNA	Initiation factors bind to ribosome
C	DNA is methylated	Poly(A) tail is added to 3' end	poly-A tail is extended	Phosphorylation of protein
D	DNA is methylated	5' capping occurs	Removal of 5' cap	Activators bind to enhancers

- 10** Young maize and wheat plants were grown to maturity at high and low temperatures. The rate of photosynthesis in each of these mature plants was measured at different temperatures. The rate of photosynthesis was measured as the amount of CO_2 used per dm^2 of leaf per hour. The results are shown in the graph below.



What information can be concluded from the graph above?

- 1 For plants grown at high temperature, the rate of photosynthesis is optimum at 25°C in maize and 18°C in wheat.
 - 2 For plants grown at high temperature, maize had a greater increase in rate of photosynthesis compared to wheat until optimum temperature was reached.
 - 3 The rate of photosynthesis was affected more significantly in maize plants than in wheat plants grown at low temperatures.
 - 4 Low temperatures slowed down the formation of membranes in maize plants but not in wheat plants.
- A** 1 and 4 only
- B** 2 only
- C** 2 and 3 only
- D** 3 and 4 only

- 11** The blue dye DCPIP can be converted to colourless DCPIP as shown below:



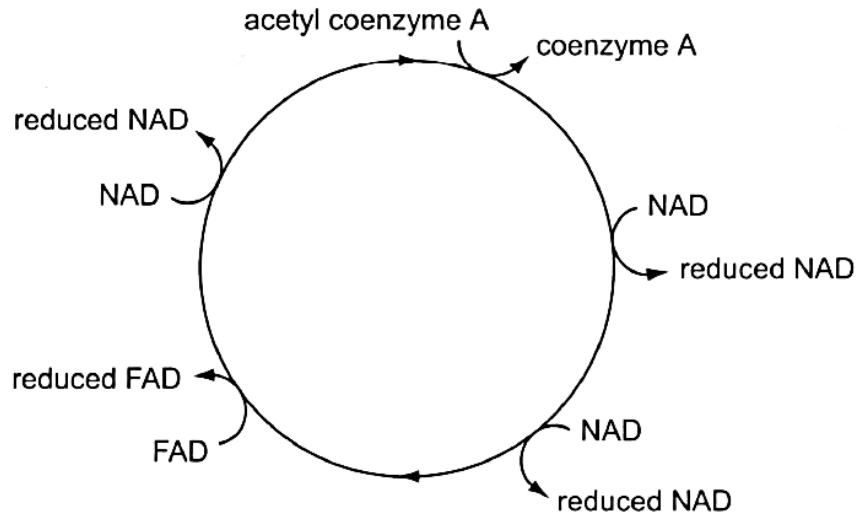
A suspension of chloroplasts was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were then prepared and treated in the following ways.

tube	contents	treatment	colour	
			at start	after 20 minutes
1	1 cm ³ chloroplast suspension + 5 cm ³ DCPIP	illuminated strongly	blue green	green
2	1 cm ³ buffer solution + 5 cm ³ DCPIP	illuminated strongly	blue	blue
3	1 cm ³ chloroplast suspension + 5 cm ³ DCPIP	left in the dark	blue green	blue green

Which statement is a possible conclusion for the observation above?

- A** Electron transfer from reduced NAD to DCPIP causes the decolourisation of DCPIP.
- B** NADP was oxidised and the electron was used to decolourise DCPIP.
- C** Light dependent reaction which occurs in the chloroplasts yield free electrons which reduced DCPIP.
- D** Either strong illumination or the buffer solution used in the extraction of chloroplasts could oxidise DCPIP.

- 12 The diagram shows the reactions of the hydrogen carriers in the Krebs cycle.



The average yield of ATP, in oxidative phosphorylation, is 3 molecules from each molecule of reduced NAD and 2 molecules from each molecule of reduced FAD.

What is the average yield of ATP, in oxidative phosphorylation, from the hydrogen carriers reduced in the Krebs cycle, from one molecule of glucose?

- A 11
 - B 22
 - C 34
 - D 36
- 13 A plant is known to be heterozygous at two gene loci. The pollen grains from this plant are used to fertilise another plant of the same genotype.

What is the probability that an embryo will be homozygous dominant at one locus?

- A 1 in 4
- B 3 in 8
- C 5 in 8
- D 1 in 16

- 14 In *Drosophila melanogaster*, the gene loci for eye colour and type of wings are next to each other on the same chromosome.

A pure-breeding fly with red eyes and long wings is crossed with a fly with pink eyes and blistered wings.

The F1 generation all had red eyes and long wings. Two of the F1 were crossed and produced 208 offspring.

Which shows the numbers of each phenotype in the F2 generation?

	phenotype			
	red eyes and long wings	red eyes and blistered wings	pink eyes and long wings	pink eyes and blistered wings
A	143	13	13	39
B	156	0	0	52
C	169	0	0	39
D	116	39	41	12

- 15 Feathers in poultry can be white or coloured and this is controlled by two genes, P/p' and Q/q. The phenotypes of offspring that are expected from mating two birds, each of which is heterozygous at both loci, are shown in the Punnett square below.

gametes	(PQ)	(Pq')	(p'Q)	(p'q)
(PQ)	white feathers	white feathers	white feathers	white feathers
(Pq')	white feathers	white feathers	white feathers	white feathers
(p'Q)	white feathers	white feathers	coloured feathers	coloured feathers
(p'q)	white feathers	white feathers	coloured feathers	white feathers

Which statement best explains the proportion of white to coloured feathers in the Punnett square?

- A Dominant epistasis in which a repressor prevents the expression of epistatic gene.
- B Dominant epistasis in which the epistatic allele is P.
- C Recessive epistasis in which colour is recessive to no colour at one allelic pair.
- D Recessive epistasis in which the epistatic allele is p'.

16 Test crossing some *Drosophila* flies gave the following results:

Red eye, normal wing	59
Purple eye, vestigial wing	43
Red eye, vestigial wing	37
Purple eye, normal wing	62

The expected ratio of phenotypes for this test cross is 1:1:1:1.

A χ^2 test was performed to test the significance of the difference between the observed and the expected results.

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad v = c - 1$$

Where Σ = 'sum of...'
 v = degrees of freedom
 c = number of classes
 O = observed 'value'
 E = expected 'value'

degrees of freedom	probability, p						
	0.50	0.20	0.10	0.05	0.02	0.01	0.001
1	0.46	1.64	2.71	3.84	5.41	6.64	10.38
2	1.39	3.22	4.61	5.99	7.82	9.21	13.82
3	2.37	4.64	6.25	7.82	9.84	11.35	16.27
4	3.36	5.59	7.78	9.49	11.67	13.28	18.47

Which combination correctly describes the result of the χ^2 test?

	Probability	Difference between observed and expected results	Loci of genes
A	< 0.05	not significant	different chromosomes
B	< 0.05	significant	same chromosomes
C	> 0.05	not significant	different chromosome
D	> 0.05	significant	same chromosome

- 17** To investigate the effect of T4 phage on two different strains of bacteria, the following steps were carried out for each of the bacterial strain.

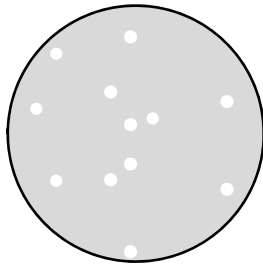
Step 1: Dilute T4 bacteriophage.

Step 2: Mix diluted bacteriophage with bacteria and agar.

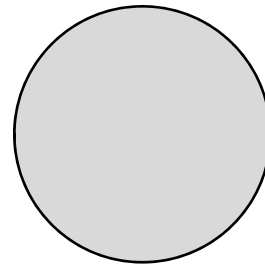
Step 3: Pour mixture from step 2 into a petri dish and allow agar to solidify.

Step 4: Incubate petri dish at 37°C for 48 hours.

It was observed that only one of the two petri dishes had plaque formation. A plaque refers to a clear and round patch of lysed bacteria cells.



Petri dish A:
a lawn of bacteria strain 1
with plaque formation



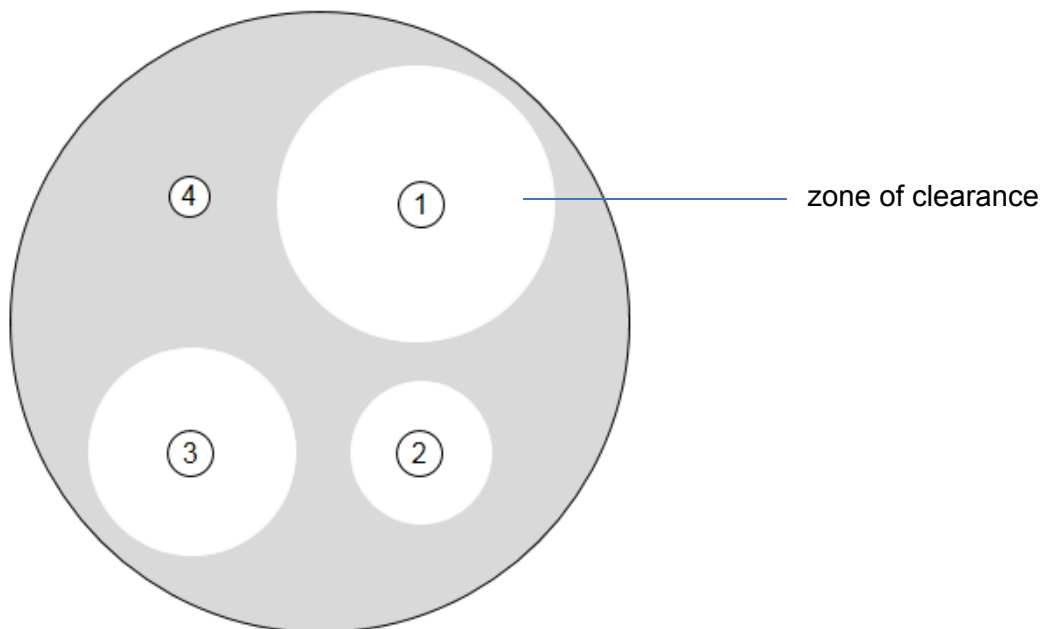
Petri dish B:
a lawn of bacteria strain 2
with no plaque formation

Which of the following can be deduced from the results?

	Petri dish A	Petri dish B
A	Bacteria strain 1 is able to kill T4 bacteriophage.	Bacteria strain 2 is not able to kill T4 bacteriophage.
B	Bacteria strain 1 is not able to kill T4 bacteriophage.	Bacteria strain 2 is able to kill T4 bacteriophage.
C	Bacteria strain 1 is resistant to T4 bacteriophage infection.	Bacteria strain 2 is not resistant to T4 bacteriophage infection.
D	Bacteria strain 1 is not resistant to T4 bacteriophage infection.	Bacteria strain 2 is resistant to T4 bacteriophage infection.

- 18** A student investigated the effect of different concentrations of unknown compound A on bacteria by looking at the diameter of the zone of clearance. The zone of clearance is an area where bacteria growth is absent on the agar plate.

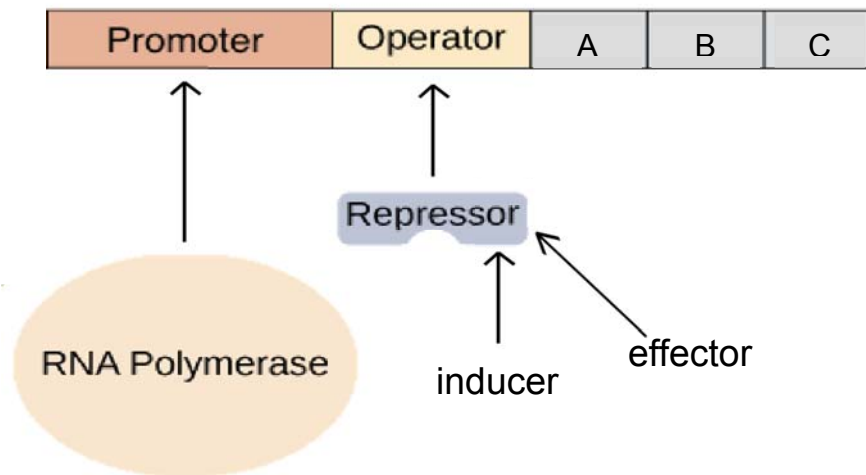
Four different concentrations of compound A are impregnated onto four separate filter paper discs and laid on top of an agar plate with bacteria. The filter paper discs are labelled 1 to 4 and the results are shown below.



Based on the results, which of the following deduction is correct?

- A** Filter paper 1 contains the lowest concentration of compound A.
- B** Filter paper 2 contains a higher concentration of compound A than filter paper 3.
- C** Filter paper 3 shows greater diffusion of compound A than that for filter paper 4.
- D** Filter paper 4 shows that compound A promotes growth of bacteria.

- 19 The diagram shows some of the molecules involved in controlling the expression of genes A, B and C in a strain of bacteria.



The following statements are with regards to the expression of genes A, B and C in the bacteria:

- RNA polymerase must bind to the promoter to transcribe structural genes A, B and C.
- The regulatory gene codes for a repressor.
- The polypeptide coded by structural gene A acts as an effector.
- The effector binds to the repressor to activate the repressor.
- The binding of inducer to repressor inactivates the repressor.

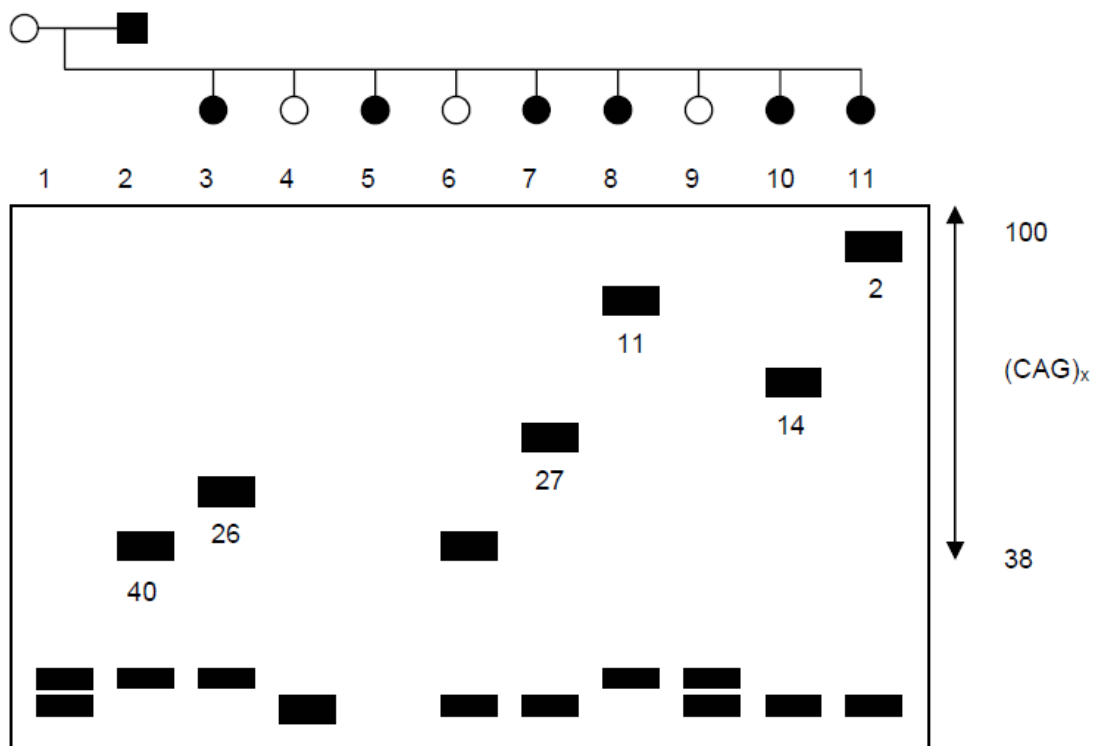
Based on the information above, which mutation does not result in constitutive expression of the structural genes?

- A** Mutation in gene A
- B** Mutation in gene B
- C** Mutation in regulatory gene
- D** Mutation in operator

For Questions **20** and **21** refer to the information below.

The inherited mutation in the Huntington's disease allele abnormally repeats the nucleotide sequence CAG more than 120 times as compared to the normal allele which has 36 CAG repeats. Using primers specific to the DNA sequence flanking this tandem repeat locus, PCR fragments of different lengths can be generated for the different alleles.

The data below shows the results of electrophoresis of the PCR fragments. The male parent, as shown by the black square box, came down with Huntington's disease when he was 40 years old. Six of his children (individuals **3, 5, 7, 8, 10, 11**) suffer from Huntington's disease as well. The age at which the symptoms first began for those suffering from Huntington's disease is shown by the number below the band from the PCR fragment.



20 What is the likely deduction for the normal individuals **4, 6, and 9**?

- A** Individuals **4, 6** and **9** have not inherited the defect causing Huntington's disease.
- B** Individuals **4, 6** and **9** do not have Huntington's disease yet but will develop Huntington's disease at some point in their lives, since the disease is inherited as a dominant trait.
- C** Only two of the three will develop Huntington's disease, since it is inherited as a dominant trait, but the data does not allow you to predict which two.
- D** Individuals **4** and **9** do not have the trait, and will not get Huntington's disease, but individual **6** is likely to start the disease when she reaches her father's age of 40.

- 21 Which of the following conclusions about Huntington's disease is valid?
- A No relationship exists between age of onset of disease and the migration rate of PCR fragments.
 - B A longer PCR fragment predicts late onset of Huntington's disease.
 - C A faster migration rate of the amplified PCR fragment predicts early onset of Huntington's disease.
 - D None of the above.

- 22 Healthy, embryonic stem cells offer great promise to the cure of many diseases because _____.

- I They are totipotent.
- II They are more readily available than other cells.
- III They can be easily cultured *in vitro*.
- IV They will differentiate in the presence of appropriate molecular signals and produce cells which are normal.

Which of the above statement(s) is/are true?

- A II, III and IV
- B I, III and IV
- C I and IV
- D IV only

- 23** Bacteria in the genus *Wolbachia* infect many butterfly species. They are passed from one generation to the next in eggs, but not in sperm, and they selectively kill developing male embryos.

In Samoa in the 1960s, the proportion of male blue moon butterflies fell to less than 1% of the population. However, by 2006, the proportion of males was almost 50% of the population.

Resistance to *Wolbachia* is the result of the dominant allele of a suppressor gene.

Which statements correctly describe the evolution of resistance to *Wolbachia* in the blue moon butterfly population?

- I *Wolbachia* acts as a selective agent.
- II The selective killing of male embryos is an example of artificial selection.
- III When infected with *Wolbachia*, male embryos that are homozygous for the recessive allele of the suppressor gene die.
- IV All male embryos that carry the dominant allele of the suppressor gene pass that allele to their offspring.
- V The frequency of the dominant allele of the suppressor gene rises in the butterfly population.

- A I and IV
- B I, III and V
- C II and III
- D II, IV and V

For Questions **24** and **25** refer to the information below.

The table shows equivalent amino acid sequences of part of a protein from four animal species.

animal species	amino acid sequence							
1	trp	met	val	glu	cys	asp	arg	leu
2	trp	val	met	glu	cys	asp	asp	ala
3	trp	val	val	glu	cys	asp	arg	leu
4	phe	trp	val	gly	cys	arg	asp	leu

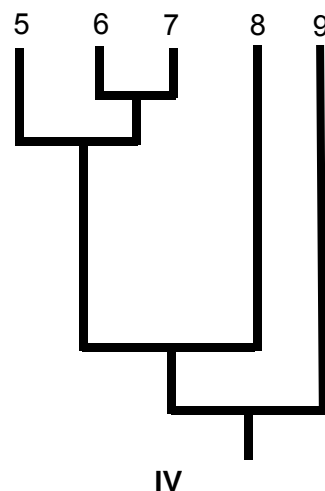
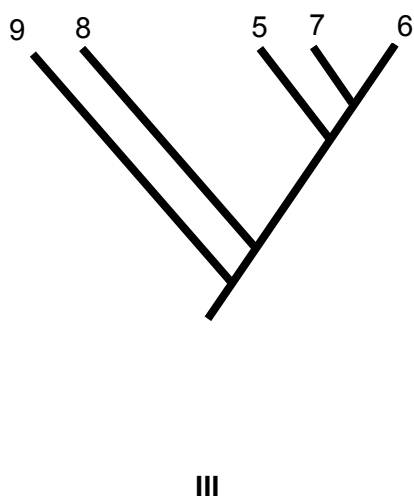
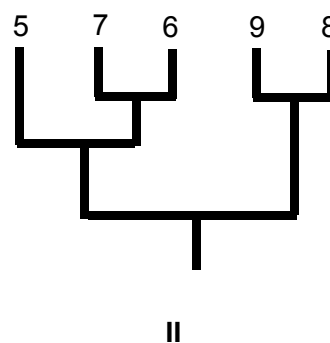
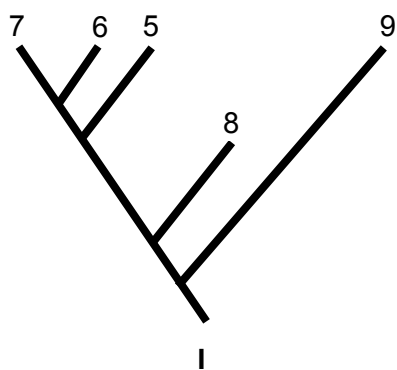
- 24** Using the technique of molecular homology, which of the following is correct?

	least closely related pair	conclusion
A	1 and 4	species 1 and 4 share the most recent common ancestor amongst the four species.
B	2 and 4	species 2 and 4 diverged the earliest amongst the four species.
C	2 and 3	species 2 and 3 share the most recent common ancestor amongst the four species.
D	3 and 4	species 3 and 4 diverged the earliest amongst the four species.

- 25 The same method was used to determine relatedness for another 5 different living species, and the amino acid differences for a particular protein are shown below.

species	5	6	7	8	9
5	0	7	8	20	23
6	7	0	3	19	22
7	8	3	0	18	21
8	20	19	18	0	20
9	23	22	21	20	0

Using the table above, which one of the following represents the probable evolutionary relationships between the species?



- A I only
 B II only
 C III only
 D III and IV

- 26 A medical technician had generated some data on the change in blood glucose concentrations of a non-diabetic and a diabetic patient at different times. Unfortunately, he had spilled some ink over the data (shown in table below) before he could analyze it.

period when the measurement was taken	blood glucose concentration/ mmol dm ⁻³	
	non-diabetic	diabetic
before a meal	4.5	4.7
1 hour after a meal	4.7	
before exercise	4.5	4.8
after prolonged exercise	4.3	

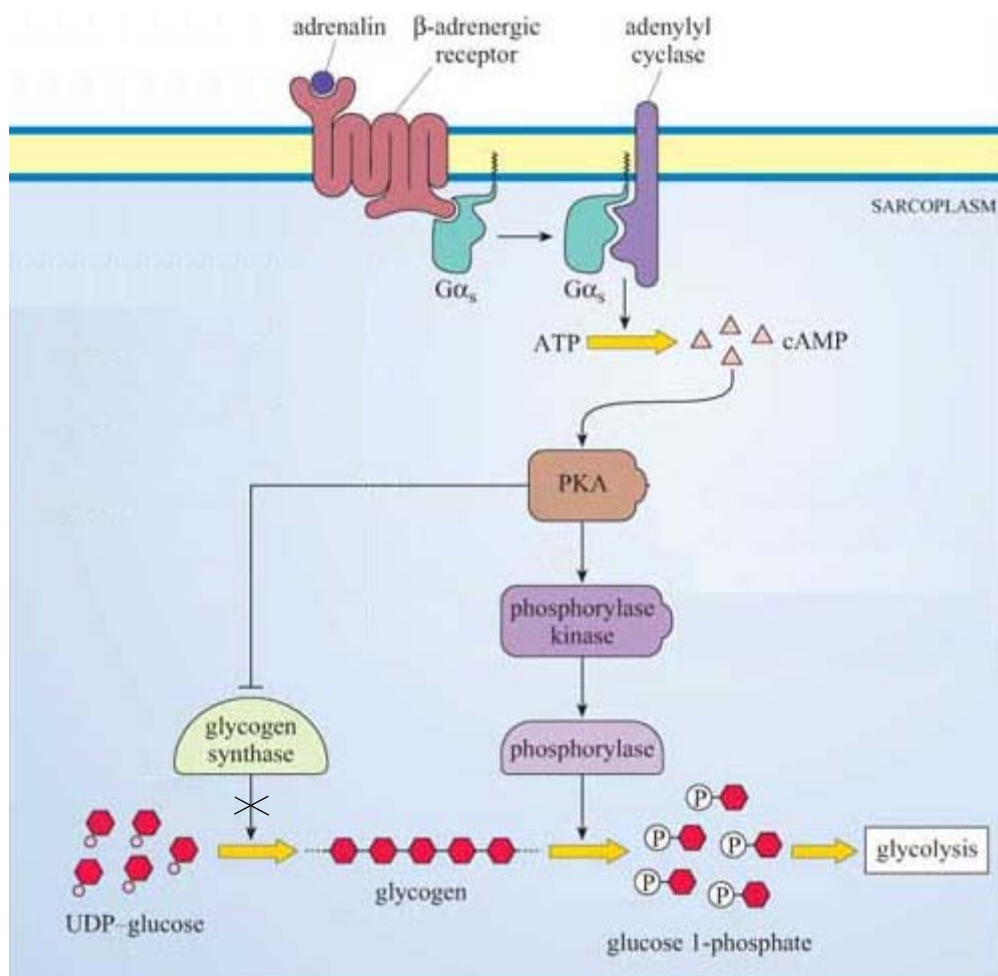
The following statements are his predicted analyses.

- I The blood glucose concentration of the diabetic patient would remain high ($>4.7\text{mmol dm}^{-3}$) 1 hour after a meal because glucose from the intestine cannot be absorbed into the blood stream.
- II The blood glucose concentration of the diabetic patient would remain high ($>4.7\text{mmol dm}^{-3}$) 1 hour after a meal because of the lack of insulin to stimulate uptake of glucose into liver cells.
- III After prolonged exercise, blood glucose concentration of the diabetic patient will remain at 4.8mmol dm^{-3} because glycogenolysis can occur.
- IV After prolonged exercise, blood glucose concentration of the diabetic patient will decrease drastically ($<4.8\text{mmol dm}^{-3}$) because there is increased rate of glucose uptake from the blood.

Which of the above predictions are correct?

- A I, and IV only
- B II and IV only
- C II, III and IV only
- D All of the above

- 27 The β -adrenergic receptor on muscle cells is a G-protein coupled receptor. Details of the cell signalling pathway in the presence of adrenalin is illustrated in the diagram below.



Which of the following is correct?

- A Inactivation of G protein occurs when GTP is displaced by GDP.
- B Signal amplification occurs at the step where G protein activates adenylyl cyclase.
- C During signal transduction, PKA can phosphorylate more than one type of molecule to activate them.
- D Muscle cells have decreased rate of glycogenesis and increased rates of glycolysis in the presence of adrenalin.

- 28** A monoclonal antibody specific for a virus was produced by identical plasma cells.

This antibody was treated with an enzyme to break the bonds between the variable and constant regions.

The separated variable and constant regions were then added to cells infected with the virus.

With regards to the experiment carried out, which statements are correct?

- I The constant regions would bind to different parts of the virus antigens.
- II The variable regions would all bind to the same part of the virus antigens.
- III The separated constant and variable regions cannot bind to the virus antigens.
- IV Phagocytes will engulf the viruses.

- A** I, II and IV
- B** I and II
- C** I and IV
- D** II only

- 29** Which method of gaining immunity can be described as natural active immunity?

- A** feeding on breast milk
- B** inhaling the chicken pox virus
- C** injection with antibodies
- D** through the placenta

- 30** Which of the following about global warming is correct?

- A** growing of crops → burning of trees to clear land → increased methane production → global warming
- B** global warming → melting of permafrost → emergence of deadly diseases → possibility of increased mortality
- C** global warming → metabolism of insects in the tropics increases → size of insects increases → amount of pesticides required to kill insect pests increases
- D** global warming → calcium carbonate uptake by corals increases → coral bleaching → disruption of food web in ocean

- END OF PAPER -

1	B	6	A	11	C	16	B	21	D	26	B
2	C	7	D	12	B	17	D	22	D	27	D
3	C	8	D	13	B	18	C	23	B	28	D
4	B	9	B	14	B	19	B	24	B	29	B
5	D	10	C	15	B	20	D	25	D	30	B